FINITE PRESENTATION MAPS ADMIT A RELATIVE NOETHERIAN STRATIFICATION OVER WHICH THE SOURCE IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 15614-15625 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 15627-15630, the complete author proof (2 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The scheme-only companion of the stratification toolkit; the S9 prerequisite is already registered, so the citation chain closes on day one.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma generic-flatness-stratification (S9 registered).

Source reference. Original lemma 0ASY: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 15566–15577. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of finite presentation between quasi-compact and quasi-separated schemes. Then there exists a $t \geq 0$ and closed subschemes*

$$S \supset S_0 \supset S_1 \supset \ldots \supset S_t = \emptyset$$

such that $S_i \to S$ is defined by a finite type ideal sheaf, $S_0 \subset S$ is a thickening, and $X \times_S (S_i \setminus S_{i+1})$ is flat over $S_i \setminus S_{i+1}$.

Proof. Apply Lemma 0ASY with $\mathcal{F} = \mathcal{O}_X$. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.